

Lab Report No. 04

Experiment Name

Implementation of Stack-Based Boundary Filling Algorithm Using OpenGL.

Objective

To implement the Stack-Based Boundary Filling Algorithm and fill a closed polygon using OpenGL.

Theory/Introduction

Boundary Fill Algorithm is a seed-fill algorithm used in computer graphics to fill the interior of a closed shape.

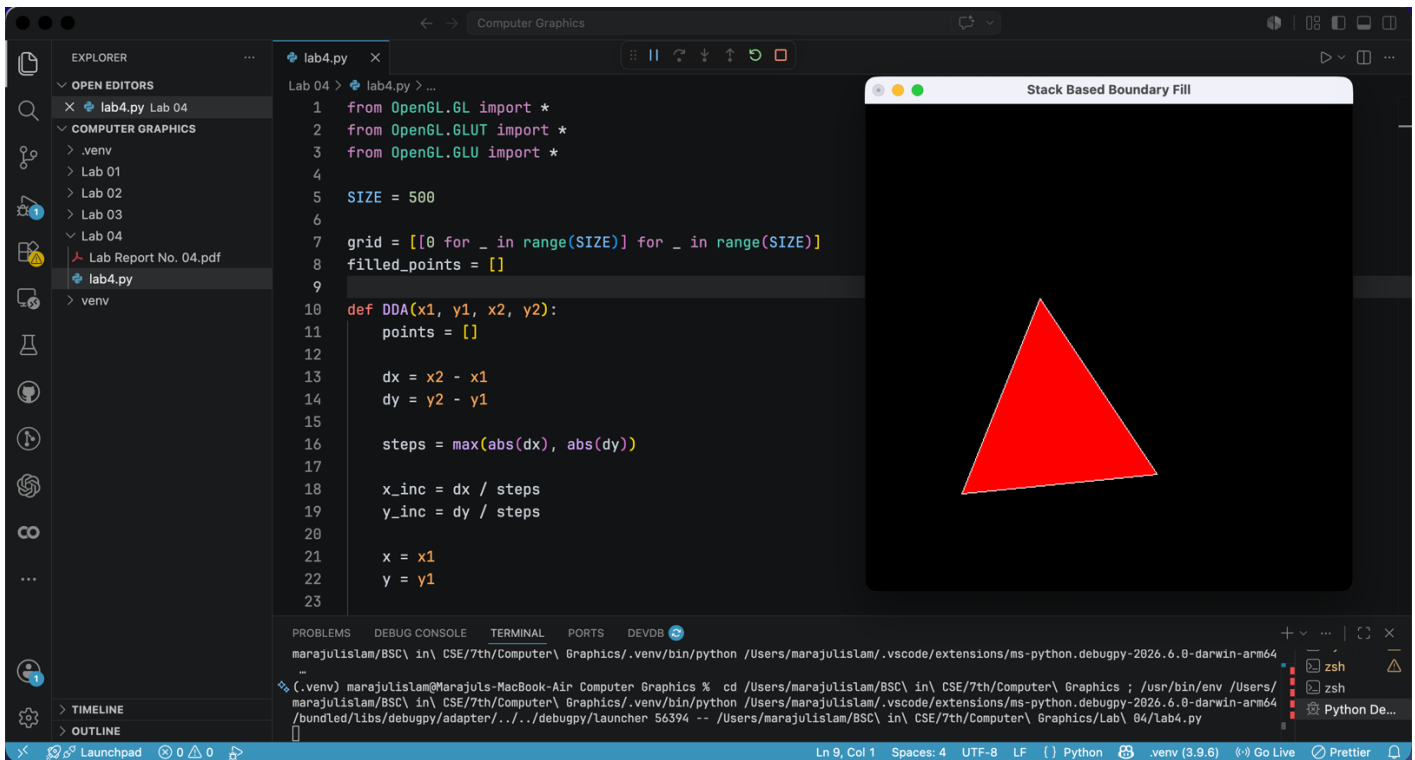
The algorithm starts from a seed point inside the boundary and fills neighboring pixels until the boundary color is reached.

In the **Stack-Based Boundary Fill Algorithm**, a stack is used instead of recursion. This avoids recursion depth limitations and is more efficient for larger areas.

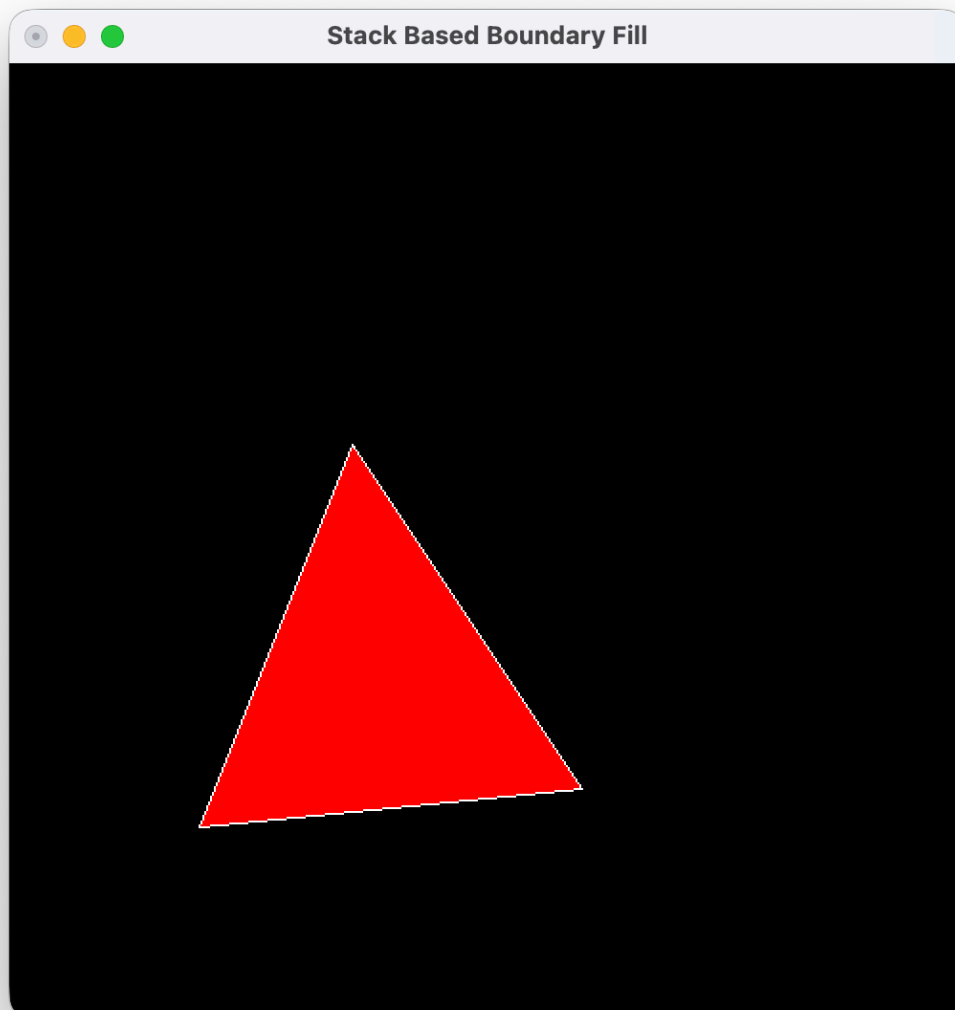
Algorithm/Procedure

1. Create a closed polygon using line drawing.
 2. Select a seed point inside the polygon.
 3. Push the seed point onto a stack.
 4. Repeat until the stack becomes empty:
 5. Pop a point from the stack.
 6. If the point is neither a boundary pixel nor already filled:
 - Fill the point.
 - Push its four neighboring pixels (up, down, left, right) onto the stack.
 7. Continue until the entire interior region is filled.
 8. Display the filled polygon.
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- Output with Code:



- Output:



Result

The Stack-Based Boundary Filling Algorithm was successfully implemented. The selected polygon was filled correctly from the seed point without crossing the boundary.

Conclusion

The experiment was completed successfully. A stack was used to fill the interior region of a closed polygon. The Stack-Based Boundary Fill Algorithm efficiently filled the area while avoiding recursion-related limitations.